

Chevalier T. Thomas Elizabeth College for Women

Title:

Evaluate chatbot conversation logs and visualize user satisfaction levels.

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ABSTRACT

This project focuses on the evaluation of chatbot conversation logs to measure user satisfaction levels using data analytics techniques. With the increasing use of chatbots in various domains such as customer support, education, and e-commerce, understanding user interaction and satisfaction has become essential. The primary objective of this project is to analyze chatbot responses and determine how effectively they meet user expectations.

The dataset used in this project consists of conversation pairs, including user messages and corresponding chatbot responses. Initially, the dataset was unstructured and required preprocessing to convert it into a usable format. The data was cleaned and transformed using Excel and Python. Sentiment analysis techniques were applied using Python libraries to classify user messages into positive, neutral, or negative categories.

To quantify user satisfaction, numerical ratings were assigned based on sentiment polarity. SQL was used to perform structured queries and aggregate analysis, enabling efficient data handling and validation. Power BI was utilized to design interactive dashboards that visually represent sentiment distribution, rating patterns, and overall chatbot performance.

The results of the analysis indicate that a majority of interactions fall under neutral and positive categories, suggesting that the chatbot performs reasonably well. However, the presence of negative sentiment highlights certain limitations in response quality. This project demonstrates the importance of data-driven approaches in evaluating and improving chatbot systems.

INTRODUCTION

In recent years, chatbots have become an integral part of digital communication systems. They are widely used in industries such as customer service, healthcare, education, and banking to automate interactions and provide instant responses. Despite their advantages, chatbots often face challenges in understanding user intent and delivering accurate responses. Therefore, evaluating chatbot performance is crucial for improving user experience.

This project aims to analyze chatbot conversation logs using data analytics tools and techniques. By examining user messages and chatbot responses, it is possible to identify patterns, trends, and areas of improvement. The use of sentiment analysis allows us to understand the emotional tone of user interactions, which directly reflects satisfaction levels.

The project integrates multiple tools, including Excel, Python, SQL, and Power BI, to perform end-to-end data analysis. Excel is used for initial data handling, Python for advanced processing and sentiment analysis, SQL for structured querying, and Power BI for visualization. This multi-tool approach ensures a comprehensive analysis of chatbot performance.

Overall, this project highlights how data analytics can be applied to real-world problems and demonstrates the importance of continuous evaluation in improving chatbot systems.

PROBLEM STATEMENT

Chatbots are designed to simulate human conversation and provide automated responses to user queries. However, many chatbots fail to deliver accurate or satisfactory responses, leading to poor user experience. The challenge lies in evaluating chatbot performance and identifying areas where improvements are required.

The main problem addressed in this project is to measure user satisfaction based on chatbot interactions. Since chatbot conversations generate large volumes of data, manual evaluation is not feasible. Therefore, a systematic approach using data analytics is required to analyze conversation logs and extract meaningful insights.

Another challenge is the unstructured nature of conversation data. User messages may contain informal language, abbreviations, and variations in expression, making analysis difficult. Additionally, there is a need to convert qualitative data (text) into quantitative metrics (ratings) for better evaluation.

This project aims to solve these challenges by applying sentiment analysis techniques, assigning ratings, and visualizing the results. By doing so, it provides a clear understanding of chatbot performance and helps in identifying areas for improvement.

TOOLS USED

Excel

Excel was used for initial data structuring, formatting, and basic analysis. Columns such as sentiment and rating were manually verified, and pivot tables were used to summarize data.

SQL (MySQL)

SQL was used to store and query the dataset. Aggregation queries such as COUNT, AVG, and GROUP BY were used to analyze sentiment distribution and average ratings. SQL helped in efficient data retrieval and validation.

Python

Python was used for advanced data analysis and sentiment classification. Libraries such as Pandas and TextBlob were used to process data and perform sentiment analysis. Matplotlib was used to generate visualizations.

Power BI

Power BI was used to create an interactive dashboard. Visualizations such as bar charts, pie charts, KPI cards, and tables were created to present insights in a clear and user-friendly manner.

DATA CLEANING PROCESS

Data cleaning is a critical step in the data analysis process, as it ensures the accuracy and reliability of results. In this project, the dataset underwent several cleaning and preprocessing steps.

Initially, the raw dataset was examined to identify missing values, inconsistencies, and formatting issues. Any incomplete or irrelevant records were removed to maintain data quality. Duplicate entries were identified and eliminated to avoid redundancy in analysis.

Text data was standardized by converting all messages to lowercase and removing unnecessary characters such as punctuation and extra spaces. This helped in improving the accuracy of sentiment analysis.

In Python, sentiment analysis was performed using the TextBlob library. Each user message was analyzed to determine its polarity, which was then used to classify the sentiment as positive, neutral, or negative. Based on this classification, a rating system was implemented to assign numerical values.

Finally, the cleaned dataset was exported as a CSV file for further analysis in SQL and Power BI. This structured and cleaned dataset ensured accurate and meaningful insights during analysis.

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand the structure and characteristics of the dataset. The primary objective of EDA is to identify patterns, trends, and relationships within the data.

The first step involved calculating basic statistics such as count, mean, and distribution of ratings. Sentiment distribution was analyzed to determine the proportion of positive, neutral, and negative interactions. It was observed that neutral interactions were the most common, followed by positive interactions.

Various visualizations were created to represent the data effectively. Bar charts were used to display sentiment counts, while pie charts showed the percentage distribution of ratings. Histograms were used to analyze the frequency of ratings.

EDA also helped in identifying outliers and unusual patterns in the data. For example, certain messages with extreme sentiment values were analyzed separately to understand their impact.

Overall, EDA provided a comprehensive understanding of the dataset and served as a foundation for further analysis and visualization.

KEY INSIGHTS

The analysis of chatbot conversation data revealed several important insights. The majority of user interactions were classified as neutral, indicating that the chatbot provides acceptable but not exceptional responses. Positive interactions suggest that the chatbot is capable of delivering satisfactory responses in certain cases.

However, the presence of negative sentiment highlights areas where the chatbot fails to meet user expectations. These negative interactions may be due to incorrect responses, lack of understanding, or incomplete answers.

The average rating calculated from the dataset indicates a moderate level of user satisfaction. This suggests that while the chatbot is functional, there is significant room for improvement.

Another key observation is the variation in user responses, which reflects diverse user expectations and communication styles. This emphasizes the need for more advanced natural language processing techniques to improve chatbot performance.

These insights provide valuable guidance for enhancing chatbot systems and improving overall user experience.

BUSINESS RECOMMENDATIONS

Based on the analysis, several recommendations can be made to improve chatbot performance. First, the chatbot should be trained with more diverse and comprehensive datasets to improve its understanding of user queries.

Second, implementing machine learning models can enhance the chatbot's ability to learn from past interactions and improve over time. Continuous feedback mechanisms should be introduced to collect user input and refine responses.

Third, the chatbot should be optimized to handle complex queries more effectively. This can be achieved by integrating advanced natural language processing techniques and context-aware responses.

Additionally, regular monitoring of chatbot performance using dashboards and analytics tools is essential. This will help in identifying issues and making data-driven improvements.

By implementing these recommendations, organizations can enhance chatbot efficiency, improve user satisfaction, and achieve better overall performance.

CONCLUSION

This project successfully demonstrated the application of data analytics techniques in evaluating chatbot performance. By analyzing conversation logs, it was possible to measure user satisfaction and identify areas for improvement.

The integration of multiple tools such as Excel, Python, SQL, and Power BI provided a comprehensive approach to data analysis. Each tool played a crucial role in different stages of the project, from data cleaning to visualization.

The findings highlight the importance of continuous evaluation and improvement of chatbot systems. While the chatbot performs reasonably well, there is potential for enhancement in response accuracy and user satisfaction.

Overall, this project emphasizes the value of data-driven decision-making and showcases how analytics can be used to improve real-world systems.